

Module 104: Official Survey Sampling & Variance Estimation Manual

1. INTRODUCTION TO OFFICIAL SURVEY DESIGN

Official large-scale socio-economic surveys conducted by the National Sample Survey Office (NSSO) utilize multi-stage stratified design to balance precision and field enumeration costs. In this design, the first stage units (FSUs) are census villages in the rural sector and Urban Frame Survey (UFS) blocks in the urban sector. The ultimate stage units (USUs) are households or enterprises.

2. STRATIFIED SAMPLING PRINCIPLES

Stratification divides a heterogeneous population into mutually exclusive and collectively exhaustive homogeneous subgroups (strata). Within each stratum, sampling is performed independently. By grouping similar units together, the sampling variance of estimates is substantially reduced compared to simple random sampling (SRS).

Key properties:

- Substratum formation based on population size, household consumption expenditure, or enterprise economic activity.
- Optimum allocation (Neyman Allocation) allocates larger sample sizes to strata with larger size and higher standard deviations.
- Proportional allocation allocates sample units in proportion to the total population of each stratum: $n_h = n * (N_h / N)$.

3. ESTIMATION PROCEDURE AND SURVEY WEIGHTS

Because sampling probabilities differ across strata and stages, sample data must be weighted to derive unbiased national aggregates.

- Design weights (Base weights) are calculated as the reciprocal of inclusion probabilities: $W_i = 1 / P_i$.
- Multiplier adjustments account for non-response and sample attrition across survey rounds.
- Ratio and regression estimators are utilized using auxiliary census information to calibrate population aggregates.

4. VARIANCE AND SAMPLING ERROR COMPUTATION

Variance estimation in multi-stage surveys uses the Ultimate Cluster Method:

$$V(\hat{Y}) = \sum_s \left[\left(\frac{n_s}{n_s - 1} \right) * \sum_i \left(\hat{Y}_{si} - \hat{Y}_s / n_s \right)^2 \right]$$

where n_s is the number of sample FSUs in stratum s , and $\hat{Y}_{si}$ is the weighted estimate from the i -th FSU. This formulation captures variance contributions from both primary and secondary sampling stages without requiring detailed cluster-level covariance calculations.

5. QUALITY METRICS: COEFFICIENT OF VARIATION (CV)

For official statistical releases, estimates with a Relative Standard Error (RSE) or Coefficient of Variation (CV) below 5% are classified as reliable for district-level policy formulation. Estimates with CV between 5% and 15% require state-level pooling, while estimates exceeding 20% must be flagged with explanatory caveats in statistical reports.